

Shuke Xiao, PhD
Postdoctoral Fellow

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EDUCATION AND TRAINING

Postdoc	Stanford University Advisor: Dr. Jonathan Z. Long	Jun 2022 – present
Ph.D.	Yale University Thesis “Drosophila Chemoreception and Threats of the Anthropocene” Advisor: Dr. John R. Carlson	Aug 2016 – May 2022
B.S.	Nanjing University Biological Sciences	Sep 2012 – Jun 2016

RESEARCH INTERESTS

Identification of novel signaling metabolites and biochemical pathways in:

- Various physiological conditions and pharmacological interventions
- Extreme physiology in non-model organisms

FELLOWSHIPS AND AWARDS

NIH K99/R00 Pathway to Independence Award, NIH	2025-2030
Wu Tsai Human Performance Fellowship, Stanford	2025-2027
American Heart Association Postdoc Fellowship	2024-2025
Stanford School of Medicine Dean's Fellowship	2023-2024
First Place Oral Presentation Award, CADA	2026
People's Choice Poster Award, Keystone - Solute Carriers	2026
Charles B. Carrington Memorial Award, Stanford Pathology	2025
Poster Award, Knight Initiative Symposium	2025
Outstanding Short Talk, Conference Travel Award, FASEB	2024
Basic Research Award, Stanford Pathology Retreat	2023
Spangler Award for outstanding PhD thesis, Yale University	2022
Conference Travel Award, Yale University	2020, 2022
Outstanding Graduate Award, Nanjing University	2016
Roche Diagnostics Scholarship, Roche	2014, 2015
China National Scholarship, Ministry of Education	2013

PUBLICATIONS

Research Articles

1. **Xiao S***, Tung AS*, Spaas J, Fu S, Chen X, Trinh TK, Moya-Garzon MD, Li VL, Cessna HT, Ito K, Reghupathy SC, Lin C, Lyu X, Raun SH, Tyner MDW, Germakovski A, Tondreau J, Yeckley M, Wei W, Howitt MR, Parker MD, Sprowl JA, Hinshaw SM, Long JZ. MCT6 is an intestinal Lac-Phe exporter required for metformin-associated weight loss. *Cell* (accepted). Preprint on bioRxiv, doi: 10.64898/2026.09.22.753556.

2. **Xiao S***, Wang M*, Martin TG*, Scott B, Fang X, Liu X, Yang Y, Fu S, Truong SD, Gugel JF, Maas GL, Mullen MP, Hill JH, Li VL, Markhard AL, Zhao M, Qi W, Reghupaty SC, Zhao M, Spaas J, Wei W, Moholdt T, Hawley JA, Voldstedlund CT, Richter EA, Chen X, Svensson KJ, Bernstein D, Leinwand LA, Xu Y, Long JZ. Python

- C.V. Shuke Xiao
 metabolomics uncovers a conserved postprandial metabolite and gut–brain feeding pathway. **Nature Metabolism**, 2026. (Cover, also in Research Briefing: Xiao S, Long JZ. Snake postprandial metabolite shows cross-species activation of hypothalamic neurons.) doi: 10.1038/s42255-026-01485-0. PMID: 41857429.
3. Fu S, Wang L, Li VL, Lyu X, Wei W, Shi X, Deng S, Barber JL, Tahir UA, Adams C, Carson A, Hidalgo B, Raffield LM, Wilson JG, Razumkov H, **Xiao S**, Spaas J, Fernandez D, Zhang T, Gerszten RE, Benson MD, Gray NS, Hinshaw SM, Long JZ. A small molecule PTER-selective inhibitor reduces food intake and body weight. **Cell Chemical Biology**, 2026. doi: 10.1016/j.chembiol.2026.02.005. PMID: 41806828.
 4. Wang Y, Zhang Y, Leung VH, Seradj SH, Sonmez U, Servin-Vences MR, **Xiao S**, Ren X, Wang L, Mishkanian SA, Kini SA, Long JZ, Lipomi DJ, Ye L, Patapoutian A. A key role of PIEZO2 mechanosensitive ion channel in adipose sensory innervation. **Cell Metabolism**, 2025. doi: 10.1016/j.cmet.2025.02.004. PMID: 40054462.
 5. Li VL, **Xiao S**, Schlosser P, Scherer N, Wiggenshorn AL, Spaas J, Tung AS, Karoly ED, Kottgen A, Long JZ. SLC17A1/3 transporters mediate renal excretion of Lac-Phe in mice and humans. **Nature Communications**, 2024. doi: 10.1038/s41467-024-51174-3. PMID: 39134528.
 6. **Xiao S**, Li VL, Lyu X, Chen X, Wei W, Abbasi F, Knowles JW, Tung AS, Deng S, Tiwari G, Shi X, Zheng S, Farrell L, Chen ZZ, Taylor KD, Guo X, Goodarzi MO, Wood AC, Chen YI, Lange LA, Rich SS, Rotter JI, Clish CB, Tahir UA, Gerszten RE, Benson MD, Long JZ. Lac-Phe mediates the effects of metformin on food intake and body weight. **Nature Metabolism**, 2024. (Highlighted in *Nature Metabolism*, *Nature Reviews Endocrinology*) doi: 10.1038/s42255-024-00999-9. PMID: 38499766.
 7. Wei W, Lyu X, Markhard AL, Fu S, Mardjuki RE, Cavanagh PE, Zeng X, Rajniak J, Lu N, **Xiao S**, Zhao M, Moya-Garzon MD, Truong SD, Chou JC, Wat LW, Chidambaranathan-Reghupaty S, Coassolo L, Xu D, Shen F, Huang W, Ramirez CB, Jang C, Li L, Svensson KJ, Fischbach MA, Long JZ. PTER is a N-acetyltaurine hydrolase that regulates feeding and obesity. **Nature**, 2024. doi: 10.1038/s41586-024-07801-6. PMID: 39112712.
 8. **Xiao S**, Baik LS, Shang X, Carlson JR. Meeting a threat of the Anthropocene: Taste avoidance of metal ions by *Drosophila*. **PNAS**, 2022. (Recommended in *Faculty Opinions*) doi: 10.1073/pnas.2204238119. PMID: 35700364
 9. **Xiao S**, Sun JS, Carlson JR. Robust olfactory responses in the absence of odorant binding proteins. **eLife**, 2019. (Recommended in *Faculty Opinions*, *F1000Prime*) doi: 10.7554/eLife.51040. PMID: 31651397
 10. Ma J, Yang Y, Fu Y, Guo F, Zhang X, **Xiao S**, Zhu W, Huang Z, Zhang J, Chen J. PIAS3-mediated feedback loops promote chronic colitis-associated malignant transformation. **Theranostics**, 2018. doi: 10.7150/thno.23046. PMID: 29896300.
 11. Fu Y, Ma J, Shi X, Song XY, Yang Y, **Xiao S**, Li J, Gu WJ, Huang Z, Zhang J, Chen J. A novel pyrazole-containing indolizine derivative suppresses NF-κB activation and protects against TNBS-induced colitis via a PPAR-dependent pathway. **Biochem Pharmacol**, 2017. doi: 10.1016/j.bcp.2017.03.013. PMID: 28336257.

Reviews

12. **Xiao S**, Markhard AL, Long JZ. Lactate as a chemical modification on proteins and metabolites. **Annual Review of Biochemistry**, 2026. doi: 10.1146/annurev-biochem-051024-015746. PMID: 42331744.
13. **Xiao S**, Li VL, Long JZ. Lac-Phe (N-lactoyl-phenylalanine). **Trends in Endocrinology and Metabolism**, 2024. (*Metabolite of the Month*). doi: 10.1016/j.tem.2024.05.007. PMID: 39137723.
14. Sun JS*, **Xiao S***, Carlson JR. The diverse small proteins called odorant-binding proteins. **Open Biology**. 2018. doi: 10.1098/rsob.180208. PMID: 30977439.

(* denotes co-first authorship)

PRESENTATIONS

Talks

1. Center for Metabolic Health Rising Stars Symposium, University of Utah
 “A coupled enzyme-transporter axis regulating N-lactoyl amino acids” Sep 2026
2. SiNaPS Seminar Series, UT Southwestern (invited speaker) Jun 2026

“Chemical Messengers, Metabolism and Physiology”

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| 3. 17 th Annual Meeting Chinese American Diabetes Association | Jun 2026 |
| “A coupled enzyme-transporter axis for lactoyl amino acids in energy homeostasis” | |
| 4. 26 th Bay Area Aging Meeting | May 2026 |
| “MCT6: the intestinal gatekeeper of metformin-induced Lac-Phe secretion and metabolic health” | |
| 5. 10 th Annual Frontiers in Diabetes Research Symposium (invited speaker) | May 2026 |
| “A coupled enzyme-transporter axis for lactoyl amino acids in energy homeostasis” | |
| 6. The Metabolic Physiology Meeting | Apr 2026 |
| “A coupled enzyme-transporter axis for lactoyl amino acids in energy homeostasis” | |
| 7. Keystone Symposium – Neurobiology of Ingestive Behavior | Apr 2025 |
| “A postprandial python metabolite mediates a gut-liver-brain feeding axis” | |
| 8. Stanford University Nutrition and Obesity Research Center | Oct 2024 |
| “A hibernation-inducible metabolite activates hepatic glucose production” | |
| 9. HHMI Janelia Junior Scientist Workshop – Brain-Body Communications | Sep 2024 |
| “Regulation of a lactate-derived appetite suppressant via reversible lactylation” | |
| 10. FASEB conference – Nutrient Sensing and Signaling Metabolism | Aug 2024 |
| “Gut-derived Lac-Phe regulates feeding and energy balance.” | |
| 11. 1 st Miami Symposium on Human Metabolism (invited speaker) | Feb 2024 |
| “Regulation of an exercise-inducible appetite suppressant” | |
| 12. 4 th Bay Area Metabolism Meeting (invited speaker) | Sep 2023 |
| “CNDP2/Lac-Phe mediates the anti-obesity effects of metformin” | |

Posters

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| 13. Keystone Symposium – Solute Carrier Transporters | Apr 2026 |
| “MCT6 mediates efflux of gut-derived lactoyl amino acids and metformin-associated weight loss” | |
| 14. 12 th Annual Symposium of the Wu Tsai Institute at Stanford | Oct 2025 |
| “Python metabolomics uncovers a conserved metabolite in the gut-brain feeding pathway” | |
| 15. Stanford Pathology Retreat | Oct 2025 |
| “A postprandial python metabolite mediates a gut-brain feeding axis” (flash talk) | |
| 16. 6 th Bay Area Metabolism Meeting | Sep 2025 |
| “A postprandial python metabolite mediates a gut-liver-brain feeding axis” (flash talk) | |
| 17. Knight Initiative Symposium | May 2025 |
| “A postprandial python metabolite mediates a gut-liver-brain feeding axis” | |
| 18. Stanford Pathology Retreat | Nov 2024 |
| “Lac-Phe mediates the anti-obesity effects of metformin” (flash talk) | |
| 19. Stanford Pathology Retreat | Nov 2023 |
| “A hibernation-induced molecule that prevents hypoglycemia in bears” (flash talk) | |
| 20. Genetics Society of America - 63 rd Annual Drosophila Meeting | Apr 2022 |
| “Meeting a threat in the Anthropocene: taste avoidance of metal ions” | |
| 21. Genetics Society of America - TAGC meeting | Apr 2020 |
| “Robust olfactory responses in the absence of odorant binding proteins” | |

PATENTS

J. Long, Y. Xu, **S. Xiao**, L. Leinwand. A postprandial snake metabolite for obesity. Stanford S25-054. Submitted.

TEACHING EXPERIENCES

Guest Lecturer

Stanford University

BIOE190 Design Thinking in Human Performance Research Spring 2026

Yale University

INP704 Comparative Neuroanatomy lab Spring 2018, 2019, 2022

Teaching Fellow

Yale University

BIOL101, Biochemistry and Biophysics Spring 2020

BIOL102, Cell Biology Spring 2019, 2020

BIOL103, Genetics and Development Spring, Fall 2018

MCDB221L, Model Organisms in Biological Research Fall 2019

Nanjing University

Molecular Biology (*flipped classroom format*) Spring 2015**MENTORING EXPERIENCES**

Rotation students

Ian Shriner (Spring 2026), Arvie Ann Violette (Winter 2024), Jeeyoon Na (Spring 2023)

Life Science Research Professional 1

Owen Jump (June 2024 – May 2025)

Graduate student

Steven Truong (2023-2024), Alan Tung (2024-2025), Sipei Fu (2026-present)

PROFESSIONAL AFFILIATIONS

American Heart Association

2024 - present

REVIEWING EXPERIENCES

Early-career reviewer for eLife

2025 - present